

NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2018 Preliminary Course Assessment Task 2

Tuesday, 12 June 2018

General instructions

- Working time – 60 minutes.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order **within** this paper and hand to examination supervisors.
- Commence each **new question** on a **NEW PAGE**. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: # EXTRA BOOKLETS USED:

Class: (please ✓)

- | | |
|----------------------------------|-----------------------------------|
| ☐ Mr Berry | ☐ Ms Lee |
| ☐ Mr Hwang | ☐ Mr Lin |
| ☐ Mr Ireland | ☐ Ms Ziazaris |
| ☐ Dr Jomaa | |

Marker's use only

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{10}$	$\overline{13}$	$\overline{15}$	$\overline{12}$	$\overline{8}$	$\overline{58}$	$\overline{100}$

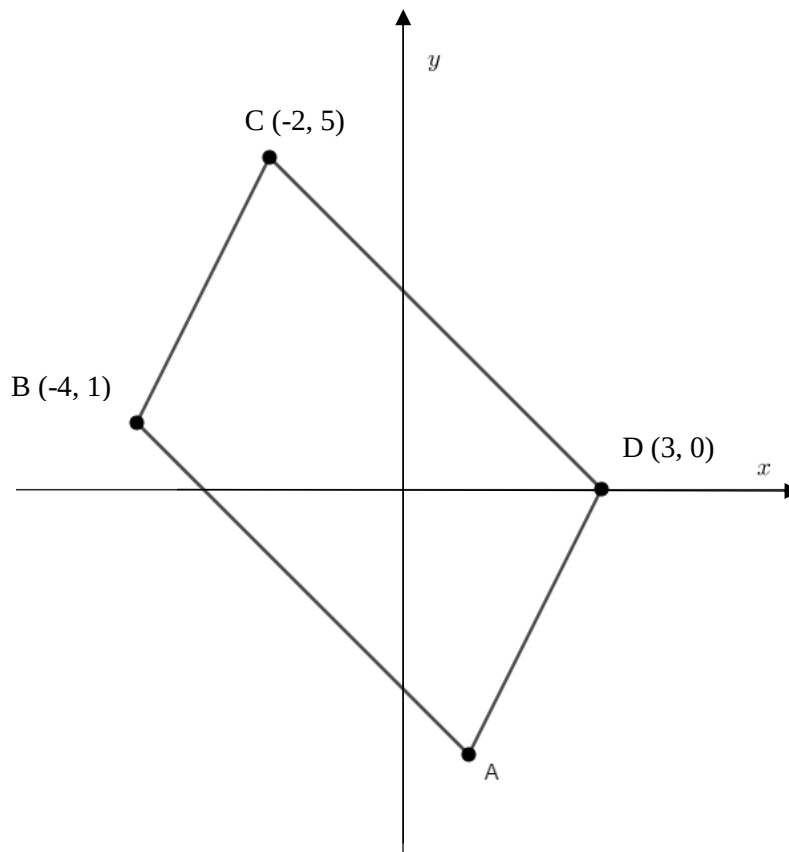
58 Marks

Attempt Question 1 to 5

Write your answers in the writing booklets supplied. Additional writing booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 1 (10 marks)

In the diagram A , $B(-4, 1)$, $C(-2, 5)$ and $D(3, 0)$ are the vertices of a parallelogram.



- | | |
|---|---|
| (a) Find the length of the interval BC . | 1 |
| (b) Show that the equation of the line through B and C is $2x - y + 9 = 0$. | 2 |
| (c) Find the angle of the inclination that CD makes with the positive direction of the x -axis. | 2 |
| (d) Find the midpoint of BD . | 1 |
| (e) Hence, or otherwise, find the co-ordinates of A . | 1 |
| (f) Find the perpendicular distance from D to BC . | 2 |
| (g) Find the area of the parallelogram $ABCD$. | 1 |

Question 2 (13 marks)

- (a) Find the acute angle between the lines $y = 3x - 5$ and $3x + 5y - 2 = 0$ giving your answer to the nearest minute.

3

- (b) The point $C(5, 3)$ divides the interval AB externally in the ratio $3:1$. If B has coordinates $(4, 1)$ find the coordinate of A .

3

- (c) Find the value of k if the points $A(-2, 8)$, $B(3, 6)$ and $C(6k - 4, 12 - 2k)$ are collinear.

2

- (d) Find the equation of the line passing through the point $(1, 2)$ and through the point of intersection of the lines $3x - 2y - 1 = 0$ and $2x + 3y - 4 = 0$.

2

- (e) Sketch the region on a number plane where the following hold simultaneously.

(Use at least a third of a page)

3

$$y \leq x^2 - 3x$$

$$y > x$$

Question 3 (15 marks)

- (a) Differentiate with respect to x and **simplify fully with common denominator where necessary.**

(i) $y = 6x\sqrt{x}$ 2

(ii) $y = (2 - 5x^3)^7$ 2

(iii) $y = -3x^2\sqrt{2x+1}$ 2

(iv) $y = \frac{5-3x}{x^2-5}$ 2

Question 3 continues on page 4

Question 3 (continued)

(b) For the curve $y = x^3 - 4x^2 + 5$,

i. Find the equation of the tangent at $x = 2$. 3

ii. Find the equation of the normal at $x = 2$. 1

(c) Differentiate the following function from **first principles**. 3

$$f(x) = 5x - x^2$$

Question 4 (12 marks)

(a) Convert $\frac{4\pi}{3}$ radians to degrees. 1

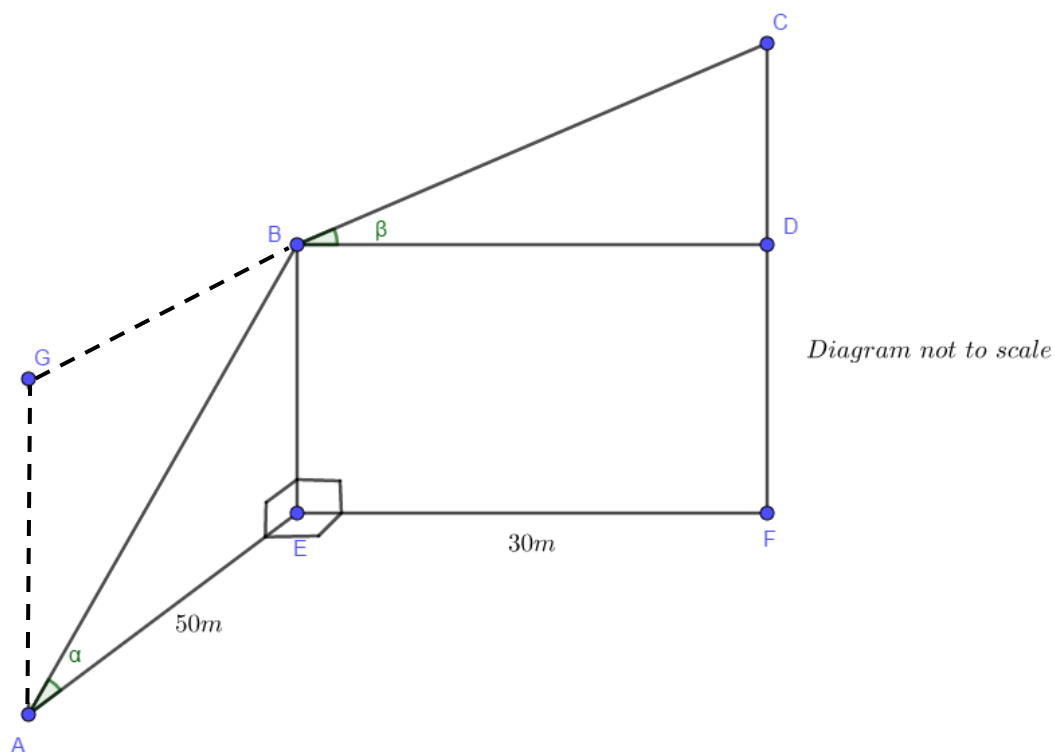
(b) Prove that $4 \sin x \cos x \cos 2x = \sin 4x$. 2

(c) (i) Write down the expansion of $\cos(x + y)$. 1

(ii) Hence find the exact value of $\cos \frac{7\pi}{12}$ as a single fraction (Do NOT rationalise the denominator) 2

(d) Given that $\tan \theta = \frac{5}{12}$, find the possible values of $\cos \frac{\theta}{2}$. 3

(e) Solve $\sqrt{3} \tan 3x = -1$, where $-\pi \leq x \leq \pi$ 3

Question 5 (8 marks)

- (a) The diagram represents part of the inside of two of the walls (built vertically and at right angles) of a parking station. AEF represents the horizontal floor. AE is 50 metres and EF is 30 metres. AB and BC represent the ramp climbing up the two walls.

The ramp climbs at angles α and β , where $\tan \alpha = \frac{1}{5}$ and $\tan \beta = \frac{1}{6}$.

- | | |
|---|---|
| (i) Find the length of FC . | 1 |
| (ii) Hence show that the distance of $AC = \sqrt{3625}$. | 2 |
| (iii) Find $\angle ABC$ to the nearest minute. | 2 |

- | | |
|---|---|
| (b) Show that $\sin 3x - \cos 3x \tan x = 2 \sin x$ | 3 |
|---|---|

End of Examination ✓

Question 1

$$\begin{aligned} \text{a) } BC &= \sqrt{(-4+2)^2 + (5-1)^2} \\ &= \sqrt{4+16} \\ &= \sqrt{20} = 2\sqrt{5} \end{aligned}$$

$$\text{b) } y - 1 = \frac{5-(1)}{-2-(-4)}(x + 4)$$

$$y = 2x + 8 + 1$$

$$2x - y + 9 = 0$$

$$2x - y + 9 = 0 \text{ as given}$$

$$\begin{aligned} \text{c) } m &= \frac{5-0}{-2-3} = \frac{5}{-5} = -1 \\ \tan^{-1}(-1) &= 135^\circ \end{aligned}$$

$$\begin{aligned} \text{d) Midpoint} &= \left(\frac{-4+3}{2}, \frac{0+1}{2} \right) \\ &= \left(-\frac{1}{2}, \frac{1}{2} \right) \end{aligned}$$

e) midpoint of $C(-2, 5)$ and $A(x, y)$
is same as midpoint of BD

$$\left(-\frac{1}{2}, \frac{1}{2} \right) = \left(\frac{-2+x}{2}, \frac{5+y}{2} \right)$$

$$\therefore x = 1, y = -4$$

$$\begin{aligned} \text{f) } \perp d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ &= \frac{|2 \times 3 + (-1) \times 0 + 9|}{\sqrt{2^2 + (-1)^2}} \\ &= \frac{|15|}{\sqrt{5}} \\ &= \frac{15\sqrt{5}}{5} = 3\sqrt{5} \text{ or } \frac{15}{\sqrt{5}} \text{ units} \end{aligned}$$

$$\begin{aligned} \text{g) } A &= b \times h \\ 2\sqrt{5} \times 3\sqrt{5} &= 30 u^2 \end{aligned}$$

Question 2

a) Angle between two lines

$$\theta = \tan^{-1} \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$m_1 = 3, m_2 = -\frac{3}{5}$$

$$\theta = \tan^{-1} \left| \frac{3 + \frac{3}{5}}{1 + 3 \times -\frac{3}{5}} \right| = \tan^{-1} \left| -\frac{9}{2} \right|$$

$$= 77^\circ 28'$$

b) $m:n = -3:1$

$$\begin{matrix} A(x, y), & B(4, 1) \\ & C(5, 3) \end{matrix}$$

$$5 = \frac{nx_1 + mx_2}{m + n}, \quad 3 = \frac{ny_1 + my_2}{m + n}$$

$$5 = \frac{1x - 3(4)}{-3 + 1}, \quad 3 = \frac{1y_1 - 3(1)}{-3 + 1}$$

$$-10 = x - 12, \quad -6 = y - 3$$

$$x = 2, y = -3$$

$$A = (2, -3)$$

c) $m_{AB} = m_{CB}$

$$\frac{6-8}{3+2} = \frac{12-2k-6}{6k-4-3}$$

$$-\frac{2}{5} = \frac{6-2k}{6k-7}$$

$$-12k + 14 = 30 - 10k$$

$$-16 = 2k$$

$$\therefore k = -8$$

$$d) 3x - 2y - 1 + k(2x + 3y - 4) = 0$$

$$3(1) - 2(2) - 1 + k(2(1) + 3(2) - 4) = 0$$

$$-2 + 4k = 0$$

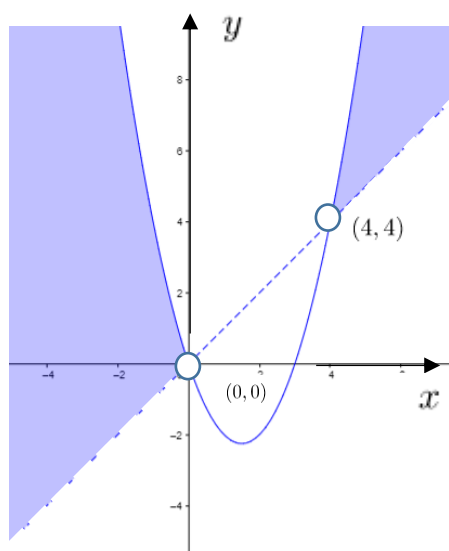
$$k = \frac{1}{2}$$

$$3x - 2y - 1 + \frac{1}{2}(2x + 3y - 4) = 0$$

$$6x - 4y - 2 + 2x + 3y - 4 = 0$$

$$8x - y - 6 = 0$$

e)



- 1- correct shape of curves
- 1- correct points of intersection
- 1- correct lines/shading

Question 3

$$a) (i) y = 6x\sqrt{x} = 6x^{\frac{3}{2}}$$

$$= 6 \times \frac{3}{2} \times x^{\frac{1}{2}}$$

$$= 9\sqrt{x}$$

$$(ii) y = (2 - 5x^3)^7$$

$$u = 2 - 5x^3$$

$$u' = -15x^2$$

$$y = 7 \times -15x^2 \times (2 - 5x^3)^6$$

$$= -105x^2(2 - 5x^3)^6$$

$$(iii) y = -3x^2\sqrt{2x+1}$$

$$u = -3x^2, \quad u' = -6x$$

$$v = \sqrt{2x+1}, \quad v' = \frac{1}{\sqrt{2x+1}}$$

$$y = uv' + vu'$$

$$y = \frac{-3x^2}{\sqrt{2x+1}} - 6x\sqrt{2x+1}$$

$$= \frac{-3x^2 - 6x(2x+1)}{\sqrt{2x+1}}$$

$$= \frac{-3x^2 - 12x^2 - 6x}{\sqrt{2x+1}}$$

$$= \frac{-15x^2 - 6x}{\sqrt{2x+1}}$$

$$= -\frac{3x(5x+2)}{\sqrt{2x+1}}$$

$$(iv) y = \frac{5-3x}{x^2-5}$$

$$u = 5 - 3x, \quad u' = -3$$

$$v = x^2 - 5, \quad v' = 2x$$

$$\frac{dy}{dx} = \frac{-3(x^2-5) - 2x(5-3x)}{(x^2-5)^2}$$

$$= \frac{3x^2 - 10x + 15}{(x^2-5)^2}$$

b)

$$(i) \frac{dy}{dx} = 3x^2 - 8x$$

$$\text{at } x = 2, \quad \frac{dy}{dx} = 3(4) - 8(2) = -4$$

Equation of tangent: $m = -4, (2, -3)$

$$y + 3 = -4(x - 2)$$

$$y = -4x + 5$$

$$(ii) m_{normal} = \frac{1}{4}, (2, -3)$$

$$y + 3 = \frac{1}{4}(x - 2)$$

$$x - 4y - 14 = 0$$

$$c) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5(x+h) - (x+h)^2 - (5x - x^2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5x + 5h - x^2 - 2xh - h^2 - 5x + x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5h - 2xh - h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(5 - 2x - h)}{h}$$

$$= 5 - 2x$$

Question 4

a)

$$\frac{4\pi}{3} \times \frac{180}{\pi}$$

$$= 240^\circ$$

$$b) 4 \sin x \cos x \cos 2x = \sin 4x$$

$$LHS = 4 \sin x \cos x \cos 2x$$

$$= 2(2 \sin x \cos x) \cos 2x$$

$$= 2 \sin 2x \cos 2x$$

$$= \sin 4x$$

$$= RHS$$

c) (i)

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

(ii)

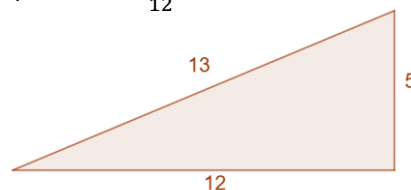
$$\cos\left(\frac{7\pi}{12}\right) = \cos\left(\frac{\pi}{3} + \frac{\pi}{4}\right)$$

$$= \cos \frac{\pi}{3} \cos \frac{\pi}{4} - \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

$$d) \tan \theta = \frac{5}{12}$$



$$\therefore \cos \theta = \pm \frac{12}{13}$$

$$\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1$$

$$\cos^2 \frac{\theta}{2} = \frac{\cos \theta + 1}{2}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{\cos \theta + 1}{2}}$$

$$= \pm \sqrt{\frac{\frac{12}{13} + 1}{2}} \text{ or } = \pm \sqrt{\frac{-\frac{12}{13} + 1}{2}}$$

$$= \pm \sqrt{\left(\frac{25}{26}\right)} \text{ or } = \pm \sqrt{\frac{1}{26}}$$

$$= \pm \frac{5}{\sqrt{26}} \text{ or } = \pm \frac{1}{\sqrt{26}}$$

$$\text{e) } \sqrt{3} \tan 3x = -1, \quad -\pi \leq x \leq \pi$$

$$\tan 3x = -\frac{1}{\sqrt{3}}, \quad -3\pi \leq 3x \leq 3\pi$$

$$(3x)_b = \frac{\pi}{6}$$

$$3x = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}, -\frac{\pi}{6},$$

$$-\pi - \frac{\pi}{6}, -2\pi - \frac{\pi}{6}$$

$$x = \frac{5\pi}{18}, \frac{11\pi}{18}, \frac{17\pi}{18}, -\frac{\pi}{18}, -\frac{7\pi}{18}, -\frac{13\pi}{18}$$

Question 5

a)

$$\tan \alpha = \frac{1}{5} = \frac{BE}{50} \quad \therefore BE = 10$$

$$\tan \beta = \frac{1}{6} = \frac{CD}{30} \quad \therefore CD = 5$$

$$\text{(i)} \quad FC = 10 + 5 = 15m$$

$$\text{(ii)} \quad FA^2 = 50^2 + 30^2$$

$$FA = \sqrt{3400}$$

$$CA^2 = 15^2 + 3400$$

$$CA = \sqrt{3625} \text{ m}$$

$$\text{(iii)} \quad AB^2 = 10^2 + 50^2$$

$$AB = \sqrt{2600}$$

$$BC^2 = 30^2 + 5^2$$

$$BC = \sqrt{925}$$

$$\cos \angle ABC = \frac{2600 + 925 - 3625}{2 \times \sqrt{2600} \sqrt{925}}$$

$$\angle ABC = 91^\circ 57'$$

$$\text{b) } \sin 3x - \cos 3x \tan x = 2 \sin x$$

$$LHS = \sin 3x - \cos 3x \tan x$$

$$= \sin 3x - \cos 3x \times \frac{\sin x}{\cos x}$$

$$= \frac{\cos x \sin 3x - \cos 3x \sin x}{\cos x}$$

$$= \frac{\sin(3x - x)}{\cos x}$$

$$= \frac{\sin 2x}{\cos x}$$

$$= \frac{2 \sin x \cos x}{\cos x}$$

$$= 2 \sin x$$

$$= RHS$$